

An Introduction to Asterisk



Telephone density in India exceeds the Internet density. Telephony, therefore, is a better means of reaching a targeted client. It certainly has greater advantages when reaching out to a largely illiterate clientele. Delve into this article if you wish to learn more about telephony and possibly do research in voice technologies.

In this article, we will learn how to install Asterisk and make an outgoing call. We will also learn how to make conference calls.

First, let us understand some basic concepts. The different methods for calling in India are:

- SIP trunk lines
- PRI lines
- GSM

SIP trunking is a Voice over Internet Protocol (VoIP) and streaming media service based on the Session Initiation Protocol (SIP). Internet telephony service providers (ITSPs) use it to deliver telephone services and unified communications to customers equipped with SIP-based private branch exchanges (IP-PBX) and unified communications facilities. Basically, trunking is transmitted voice over data. Later, we will learn how to create a SIP user and make an IVR call based on a SIP line. In India, Tata Teleservices Ltd and Reliance provide SIP trunk lines.

PRI lines are primary interface lines. Even though there is only one line that physically terminates in the customer PBX, a PRI line can receive/send 30 calls simultaneously! A PRI line is an end-to-end digital circuit.

There is also the **GSM module** for calling with the W400 GSM card, which is used to make calls from a GSM SIM card. From one SIM card, you can make five calls, simultaneously.

Installing Asterisk

I am using the Centos 6.7 (64-bit) OS, with the Sangoma PRI card A102E with echo cancellation (see Figure 1 for the Sangoma PRI card). I have placed this card in the PCI Express slot in the server.

Before installing Asterisk and the Sangoma driver, please install Centos 6.7, 64-bit. Now, run the command `yum -y kernel*`. Download `libpri` by running:

```
wget http://downloads.asterisk.org/pub/telephony/libpri/old/libpri-1.4.12.tar.gz
```

Download the complete Dahdi Linux by running the following code:

```
wget http://downloads.asterisk.org/pub/telephony/dahdi-linux-complete/releases/dahdi-linux-complete-2.6.0+2.6.0.tar.gz"
```